

Inference

Logistics

- + HW 5 due Friday
- + No homework due next week
- + Exam 1: Wednesday 10/7 in class
 - + Covers material through Friday 10/2
 - + Some review questions posted on course website
 - + You do *not* need to write any R code, but you do need to read some R output

Last time

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

$$H_0 : \beta_2 = 0 \quad H_A : \beta_2 \neq 0$$

A new research question

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Research question: Is there a relationship between school rank and grad school acceptance, after accounting for GPA and GRE score?

What are my hypotheses for this new research question?

Testing multiple parameters

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

How do we test multiple parameters in a *linear* regression model?

nested test (compare full and reduced models)
 ↙ ↘
all explanatory variables remove rank

Nested models

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

Full model (H_A):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

Reduced model (H_0):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i$$

Comparing nested models

Full model (H_A):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

Reduced model (H_0):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i$$

How should we compare these two nested models?

Comparing nested models

Deviance: for a logistic regression model, deviance is $-2 \log L$

Idea:

- + Maximizing likelihood (how we fit the model) is equivalent to minimizing deviance (like minimizing SSE in linear regression!)
- + Smaller deviance implies better fit
- + Is there a big difference in the deviance (model fit) between these models??

Change in deviance:

$$G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$$

$$G \geq 0$$

(adding variables to model can't make deviance worse — deviance can only get smaller if we add more variables)

Comparing deviance

```
m_full <- glm(admit ~ gpa + gre + as.factor(rank),  
              family = binomial, data = grad_app)  
  
m_reduced <- glm(admit ~ gpa + gre,  
                 family = binomial, data = grad_app)  
  
m_reduced$deviance - m_full$deviance
```

```
## [1] 21.82649
```

How do I know whether this is a big change in deviance? That is: is a change in deviance of **21.83** unusual under the null hypothesis?

Parametric bootstrap

- ① Fit reduced model
- ② simulate response from reduced model
(No guaranteed true for simulated data!.)
eg: $\text{Admit}_i^* \sim \text{Bernoulli}(\hat{\pi}_{i(\text{reduced})})$
- ③ Refit full model and reduced model on simulated data $\rightarrow$ deviance full (sim), deviance reduced (sim)
- ④ Recompute test statistic:
 $\text{deviance reduced (sim)} - \text{deviance full (sim)}$
- ⑤ Repeat many times!

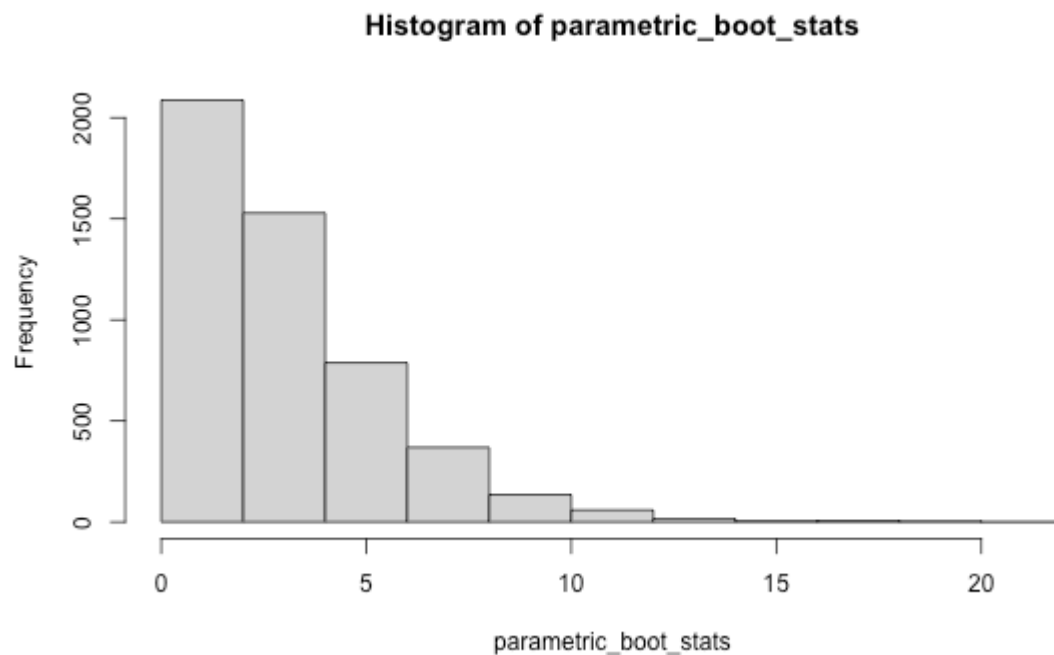
Class activity

- + Work on the class activity (linked on Canvas and course website)
- + Submit rendered HTML on Canvas at the end of class

Activity

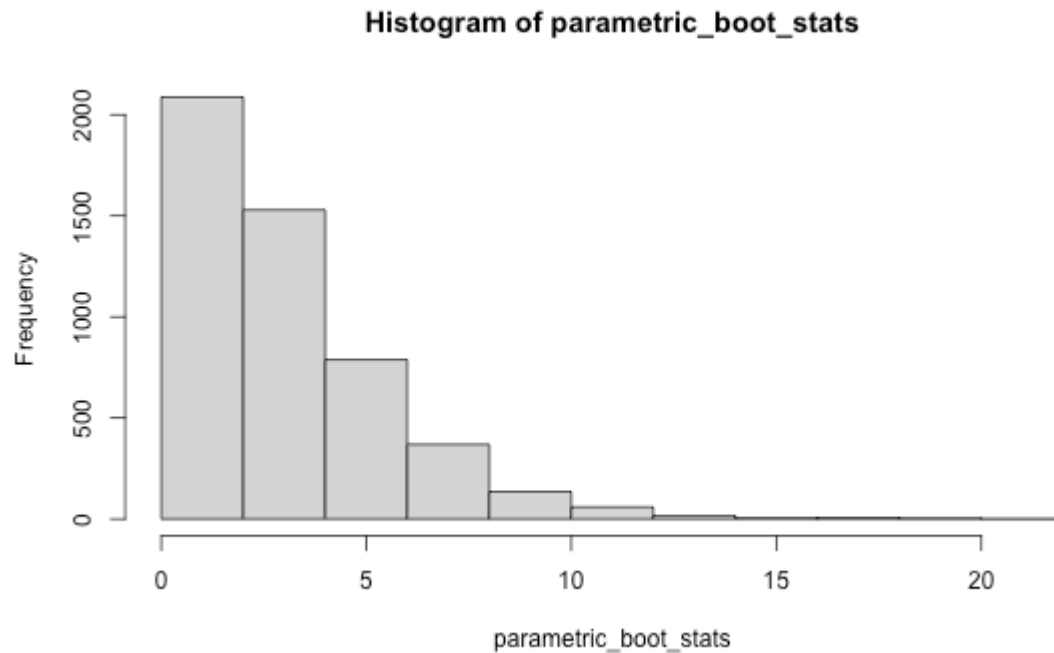
What does this distribution look like?

Activity



How unusual is the observed test statistic of 21.83, compared to this null distribution?

p-value



```
mean(parametric_boot_stats > 21.83)
```

```
## [1] 0
```